

```
In [1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

sns.set_style("whitegrid")
```

```
In [2]: df = pd.read_csv("train.csv")
df.head()
```

Out[2]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500

```
In [3]: df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
#   Column      Non-Null Count  Dtype
---  -
0   PassengerId  891 non-null    int64
1   Survived     891 non-null    int64
2   Pclass       891 non-null    int64
3   Name         891 non-null    str
4   Sex          891 non-null    str
5   Age          714 non-null    float64
6   SibSp        891 non-null    int64
7   Parch        891 non-null    int64
8   Ticket       891 non-null    str
9   Fare         891 non-null    float64
10  Cabin        204 non-null    str
11  Embarked     889 non-null    str
dtypes: float64(2), int64(5), str(5)
memory usage: 83.7 KB
```

```
In [4]: df.describe()
```

Out[4]:

	PassengerId	Survived	Pclass	Age	SibSp	Parch	Fare
count	891.000000	891.000000	891.000000	714.000000	891.000000	891.000000	891.000000
mean	446.000000	0.383838	2.308642	29.699118	0.523008	0.381594	32.204208
std	257.353842	0.486592	0.836071	14.526497	1.102743	0.806057	49.693429
min	1.000000	0.000000	1.000000	0.420000	0.000000	0.000000	0.000000
25%	223.500000	0.000000	2.000000	20.125000	0.000000	0.000000	7.910400
50%	446.000000	0.000000	3.000000	28.000000	0.000000	0.000000	14.454200
75%	668.500000	1.000000	3.000000	38.000000	1.000000	0.000000	31.000000
max	891.000000	1.000000	3.000000	80.000000	8.000000	6.000000	512.329200

```
In [5]: df.isnull().sum()
```

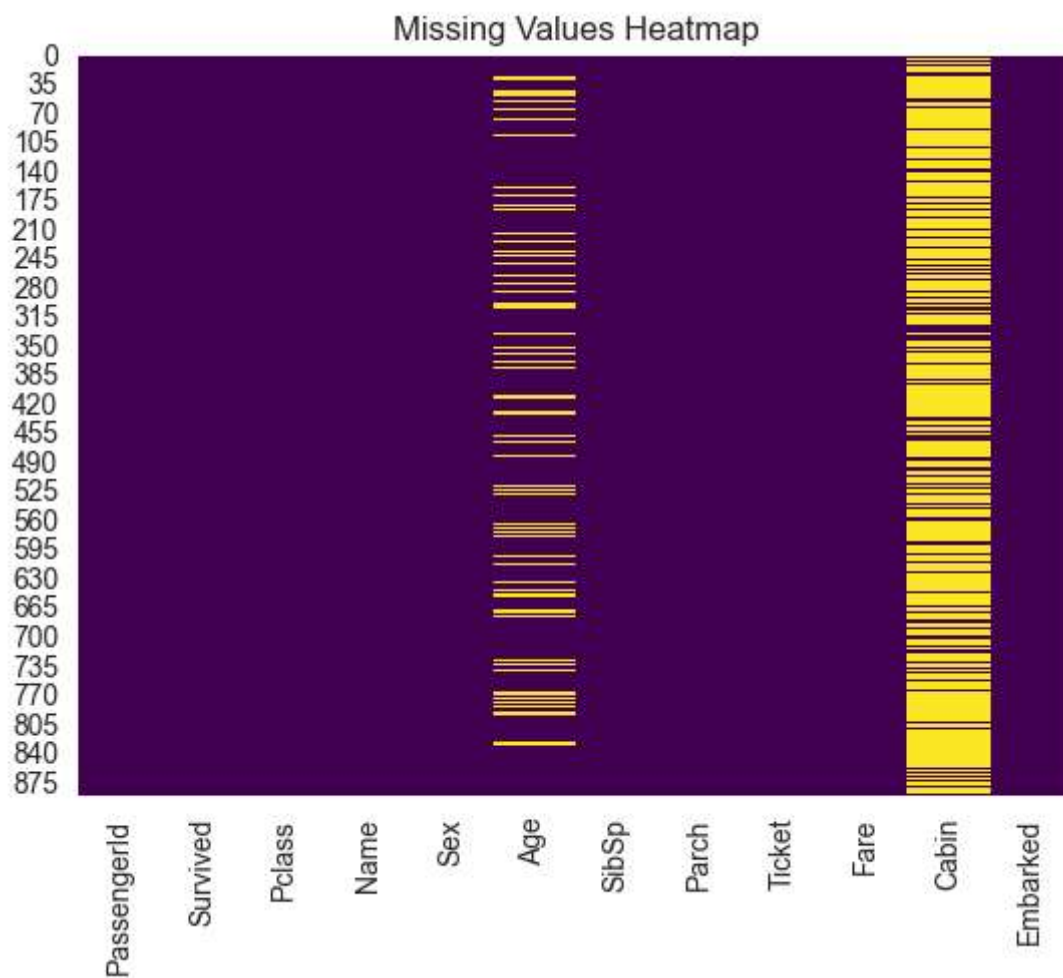
Out[5]:

PassengerId	0
Survived	0
Pclass	0
Name	0
Sex	0
Age	177
SibSp	0
Parch	0
Ticket	0
Fare	0
Cabin	687
Embarked	2
dtype: int64	

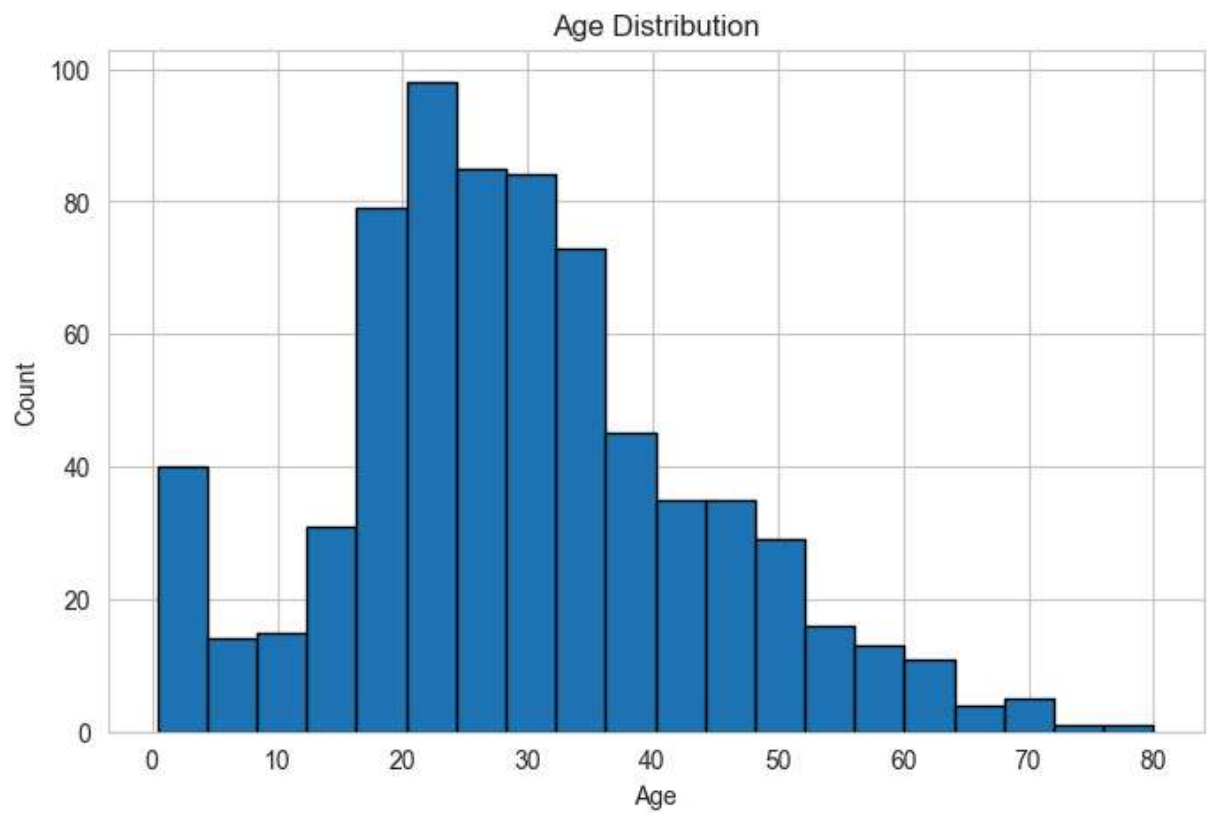
```
In [6]: df["Survived"].value_counts()
```

```
Out[6]: Survived
0      549
1      342
Name: count, dtype: int64
```

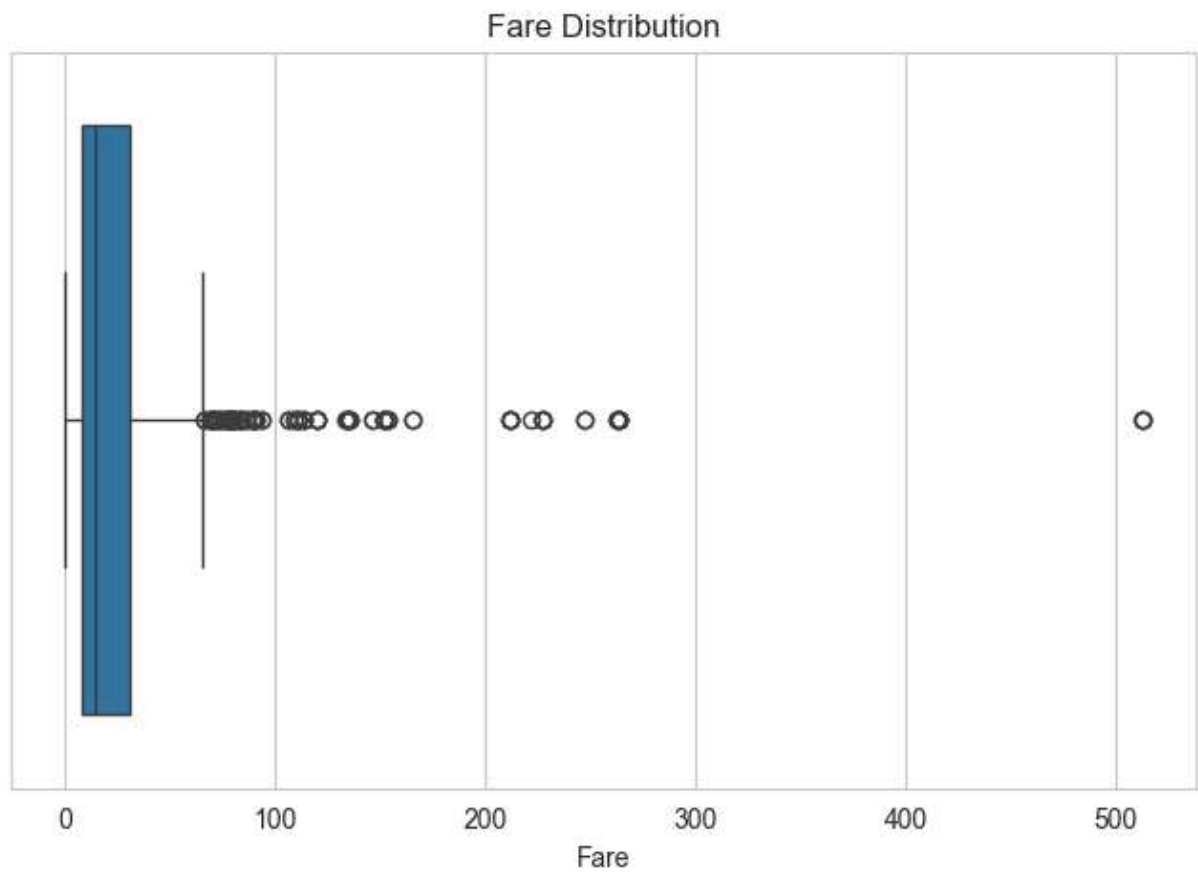
```
In [7]: sns.heatmap(df.isnull(), cbar=False, cmap="viridis")
plt.title("Missing Values Heatmap")
plt.show()
```



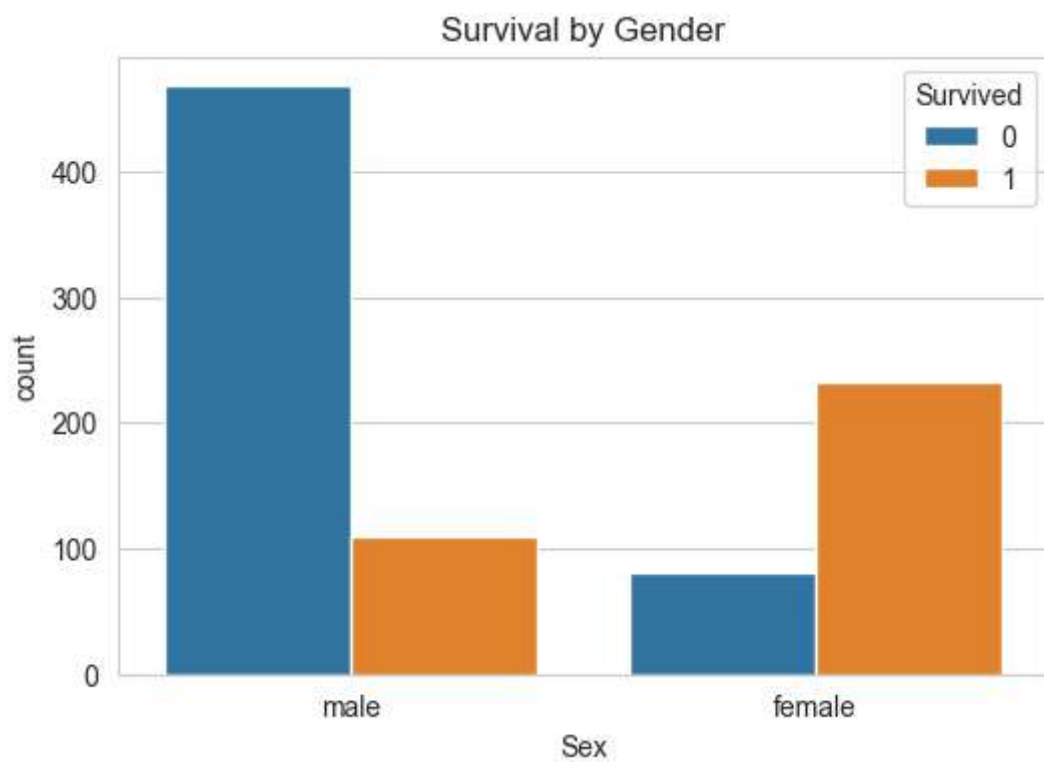
```
In [8]: plt.figure(figsize=(8,5))
plt.hist(df["Age"].dropna(), bins=20, edgecolor="black")
plt.title("Age Distribution")
plt.xlabel("Age")
plt.ylabel("Count")
plt.show()
```



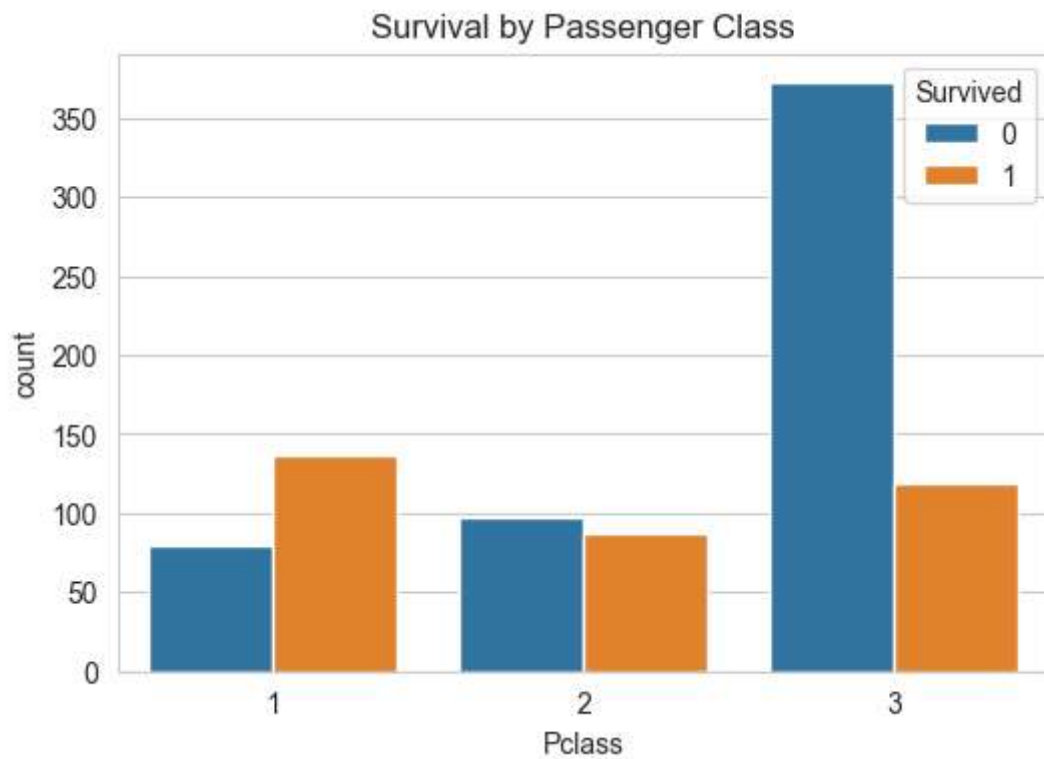
```
In [9]: plt.figure(figsize=(8,5))
sns.boxplot(x=df["Fare"])
plt.title("Fare Distribution")
plt.show()
```



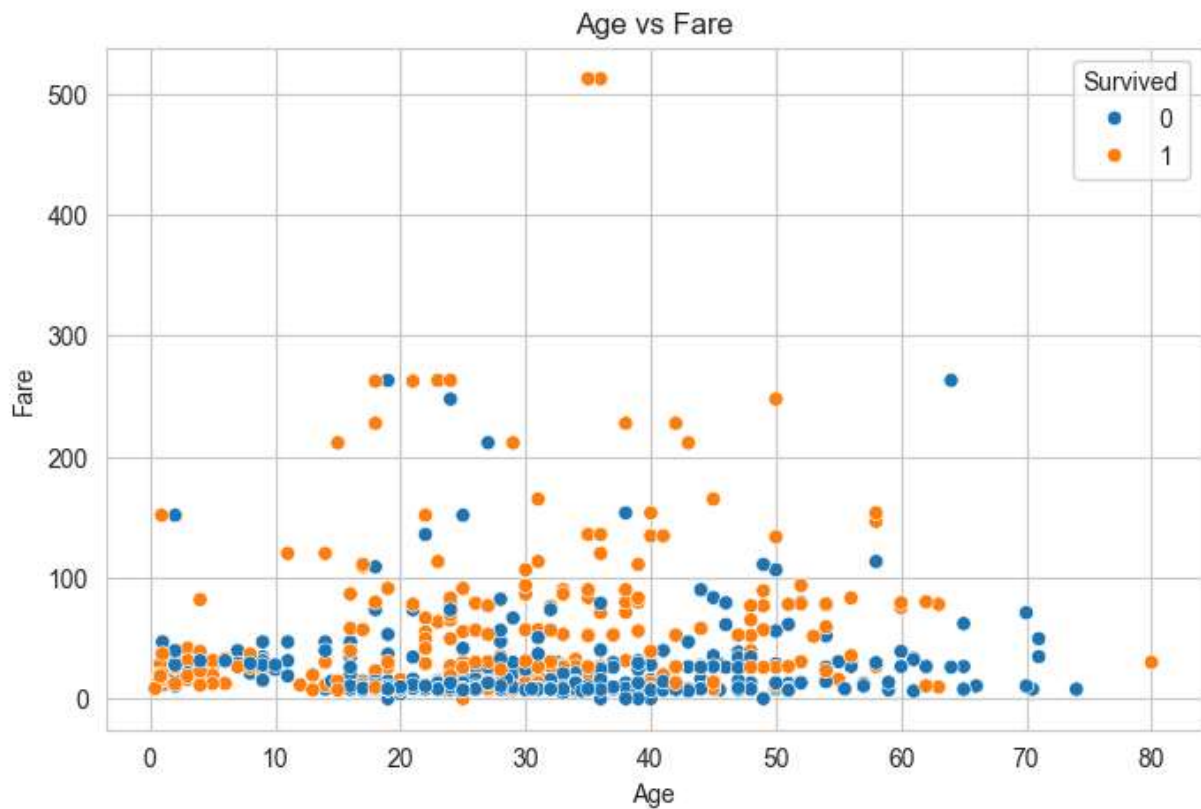
```
In [10]: plt.figure(figsize=(6,4))
sns.countplot(x="Sex", hue="Survived", data=df)
plt.title("Survival by Gender")
plt.show()
```



```
In [11]: plt.figure(figsize=(6,4))
sns.countplot(x="Pclass", hue="Survived", data=df)
plt.title("Survival by Passenger Class")
plt.show()
```



```
In [12]: plt.figure(figsize=(8,5))
sns.scatterplot(x="Age", y="Fare", hue="Survived", data=df)
plt.title("Age vs Fare")
plt.show()
```

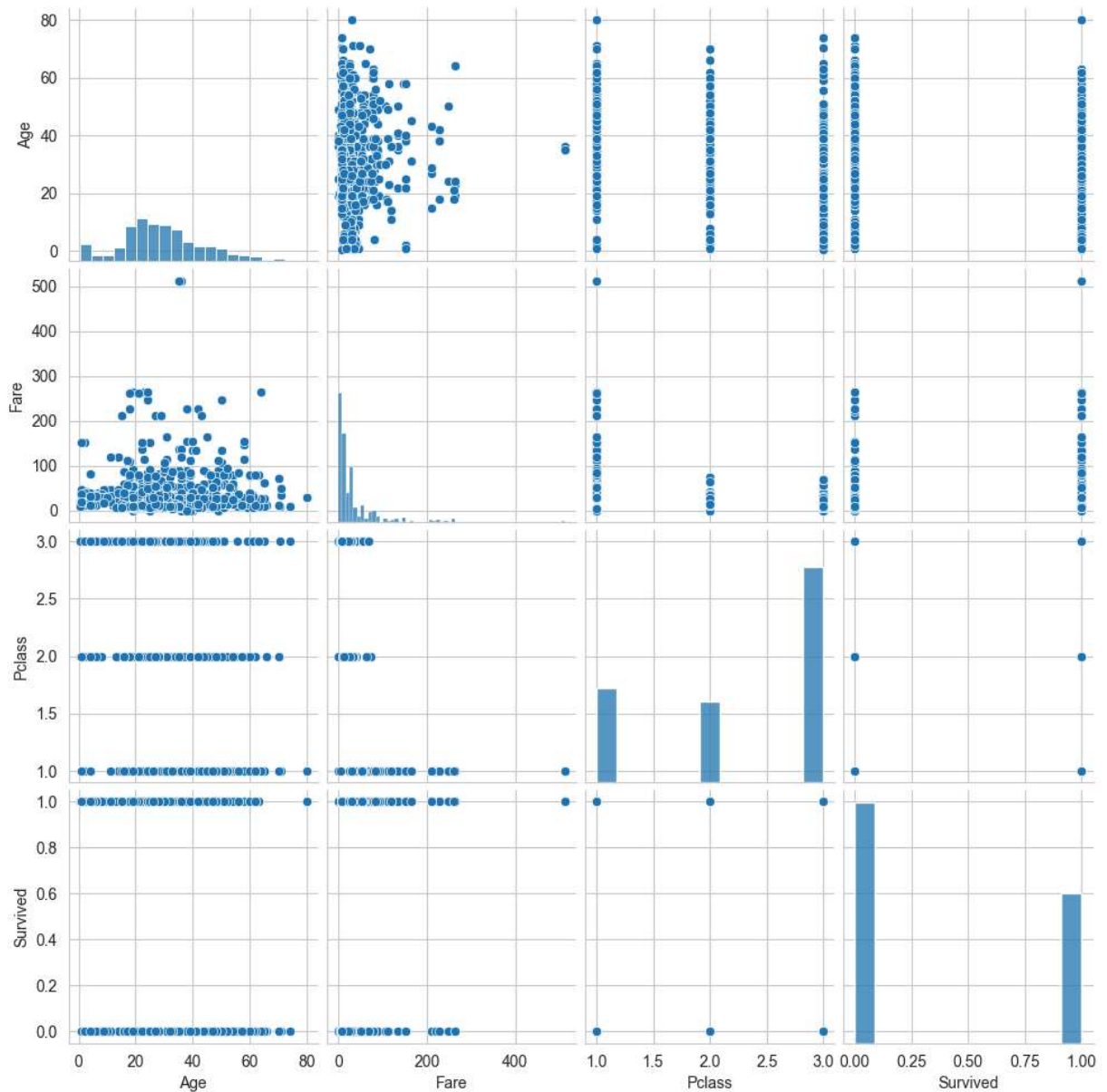


```
In [13]: df_encoded = df.copy()
df_encoded["Sex"] = df_encoded["Sex"].map({"male": 0, "female": 1})

plt.figure(figsize=(10,6))
sns.heatmap(df_encoded.corr(numeric_only=True), annot=True, cmap="coolwarm")
plt.title("Correlation Heatmap")
plt.show()
```



```
In [14]: sns.pairplot(df[["Age", "Fare", "Pclass", "Survived"]])
plt.show()
```



```
In [ ]: ## Titanic EDA Summary
```

- The dataset contains passenger details like age, sex, class, fare, and survival.
- Many missing values exist in Age and Cabin columns.
- Females had a higher survival rate compared to males.
- First-class passengers had better chances of survival.
- Fare shows outliers indicating premium tickets.
- Passenger class and gender strongly influenced survival.

```
In [ ]:
```

```
In [ ]:
```